

COMPANION SLIDES

Chapter 6

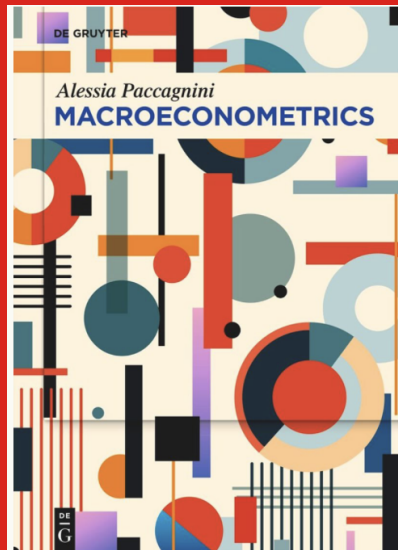
DSGE Models

From RBC foundations to HANK

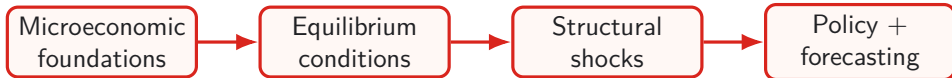
Alessia Paccagnini

Macroeconometrics – De Gruyter, 2026

DOI: 10.1515/9783111572802



Build macroeconomic dynamics from
optimizing agents + equilibrium + shocks.



- ▶ Stylized facts of business cycles
- ▶ Basic neoclassical / RBC model
- ▶ Steady state and log-linearization
- ▶ Blanchard–Kahn conditions
- ▶ Smets–Wouters model
- ▶ Calibration and moments matching
- ▶ Bayesian DSGE estimation
- ▶ Advanced MCMC and nonlinear DSGEs
- ▶ Model comparison and alternative estimation
- ▶ DSGE–VAR / DSGE–FAVAR and misspecification
- ▶ Financial frictions
- ▶ Real-time data and imperfect information
- ▶ TANK, HANK and further heterogeneity

- ▶ Aggregate dynamics are derived from explicit household and firm optimization.
- ▶ Structural parameters connect behavior to preferences, technology and constraints.
- ▶ Policy experiments can be interpreted through equilibrium responses rather than reduced-form correlations alone.
- ▶ The framework is designed to confront the **Lucas critique**: policy changes may alter reduced-form relationships.
- ▶ DSGEs can be used for shock analysis, forecasting and counterfactual policy exercises.

A model needs empirical targets

Before constructing the theory, document volatility, persistence and comovement of key macroeconomic variables.

- ▶ Consumption is typically smoother than output.
- ▶ Investment is substantially more volatile than output.
- ▶ Output, hours and investment are procyclical.
- ▶ Business-cycle fluctuations are persistent.

Key takeaway

Stylized facts are benchmarks against which the model's quantitative performance is evaluated.

Household

Chooses consumption and labor to maximize expected lifetime utility subject to a resource constraint.

Firm

Produces output using capital, labor and stochastic technology.

A compact production side is

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}, \quad \log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A.$$

- ▶ Household optimality delivers an **Euler equation**: consumption today versus tomorrow.
- ▶ Intratemporal optimality links the marginal utility of consumption to the disutility of labor.
- ▶ Competitive firms equate factor prices to marginal products.
- ▶ Capital evolves through investment and depreciation.
- ▶ Market clearing closes the model.

DSGE logic

Individual optimality conditions + technology + market clearing $\Rightarrow$ a system of dynamic equilibrium equations.

Steady state

Find constant values of endogenous variables when shocks are at their means. This provides the point around which the model is approximated.

Log-linearization

Express variables as deviations from steady state and obtain a linear rational-expectations system that can be solved with standard methods.

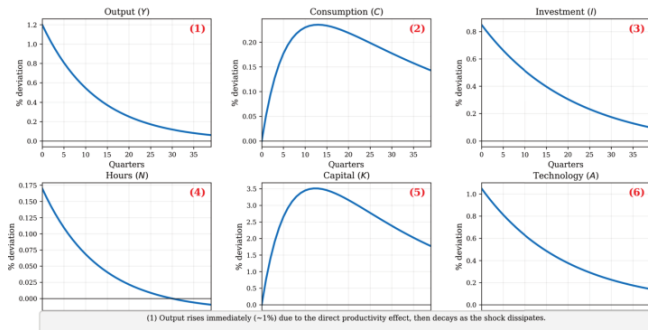
$$\mathbb{E}_t \begin{bmatrix} s_{t+1} \\ c_{t+1} \end{bmatrix} = A \begin{bmatrix} s_t \\ c_t \end{bmatrix} + B\varepsilon_t.$$

Existence and uniqueness

A unique stable solution requires the number of eigenvalues outside the unit circle to equal the number of forward-looking variables.

- ▶ Too few unstable roots: indeterminacy.
- ▶ Too many unstable roots: no stable rational-expectations equilibrium.
- ▶ In monetary models, the Taylor principle is closely related to determinacy.

Impulse Responses to a One Percent Technology Shock



- ▶ A positive technology shock raises output immediately.
- ▶ Consumption responds more smoothly.
- ▶ Investment reacts strongly.
- ▶ Capital accumulates gradually, creating endogenous persistence.

Why enrich the baseline?

A technology-only flexible-price model is too restrictive for many policy questions.

- ▶ Add nominal price and wage rigidities.
- ▶ Add habit formation and investment adjustment costs.
- ▶ Add monetary policy through an interest-rate rule.
- ▶ Introduce multiple structural shocks.
- ▶ Preserve structural interpretation while improving empirical fit.

Key frictions

- ▶ sticky prices and wages;
- ▶ habit formation;
- ▶ investment adjustment costs;
- ▶ interest-rate smoothing.

Key shocks

- ▶ technology;
- ▶ risk premium;
- ▶ investment-specific;
- ▶ monetary policy;
- ▶ government spending;
- ▶ price and wage markups.

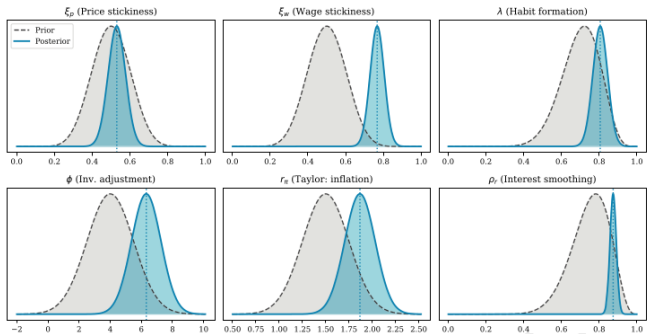
Key takeaway

Medium-scale DSGEs combine economic structure with enough frictions and shocks to confront macroeconomic data.

	Calibration	Estimation
Parameter choice	micro evidence / long-run targets	likelihood + data
Uncertainty	often limited	explicit sampling/posterior uncertainty
Strength	transparent discipline	formal inference
Challenge	parameter choices can be debated	computational and identification issues

The chapter treats calibration and formal estimation as complementary tools rather than mutually exclusive philosophies.

1. Solve the DSGE model for a candidate parameter vector θ .
2. Write the solution in state-space form.
3. Evaluate $p(y \mid \theta)$, typically with the Kalman filter for linear-Gaussian models.
4. Combine likelihood and prior: $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$.
5. Simulate the posterior with Metropolis–Hastings or related methods.
6. Use posterior draws for IRFs, shock decompositions and policy analysis.



The Smets–Wouters application illustrates how the data update prior beliefs about price and wage stickiness, habit formation, investment adjustment costs, the Taylor-rule response and interest-rate smoothing.

- ▶ Standard random-walk Metropolis–Hastings can mix slowly in high-dimensional or irregular posteriors.
- ▶ Randomized blocking can improve exploration of correlated parameter spaces.
- ▶ Tailored proposal densities use local posterior curvature to improve efficiency.
- ▶ Multimodality and weak identification require special care.

Example in the chapter

The Tailored Randomized Block Metropolis–Hastings approach of Chib and Ramamurthy is discussed as an efficient alternative for complex DSGE posteriors.

When the Kalman filter is no longer enough

Higher-order solutions, nonlinearities and occasionally binding constraints can imply non-Gaussian state dynamics.

- ▶ Particle filters / sequential Monte Carlo can approximate the likelihood.
- ▶ Second-order perturbation captures risk and precautionary effects absent at first order.
- ▶ Strong nonlinearities may require projection or value-function methods.

Bayesian comparison

Marginal likelihoods and Bayes factors compare competing DSGE specifications while integrating parameter uncertainty.

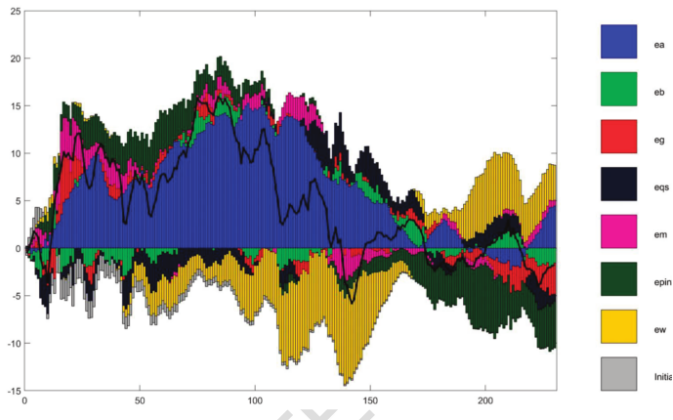
Other approaches

The chapter also discusses GMM, simulation-based methods and impulse-response matching.

Key takeaway

A structurally appealing model still has to earn empirical support.

- ▶ DSGEs impose strong cross-equation restrictions derived from theory.
- ▶ VARs are more flexible but offer less structural discipline without identification.
- ▶ DSGE–VARs introduce a controlled amount of deviation from DSGE restrictions.
- ▶ DSGE–FAVARs expand the information set using latent factors.
- ▶ Misspecification diagnostics ask where theory and data disagree rather than assuming the structural model is exact.



An estimated DSGE can decompose observed macroeconomic fluctuations into contributions from structural shocks.

Interpretation

This connects the model's structural disturbances to historical episodes in the data.

Financial accelerator

Borrower balance sheets and the external finance premium can amplify and propagate otherwise modest shocks.

- ▶ Lower net worth raises external financing costs.
- ▶ Tighter financing reduces investment and activity.
- ▶ Falling activity can further weaken balance sheets: an amplification loop.
- ▶ Banking frictions extend the mechanism to intermediaries' capital and lending capacity.

- ▶ Policymakers observe noisy and subsequently revised data.
- ▶ Agents may receive signals about future fundamentals rather than observe them perfectly.
- ▶ Distinguishing **news** from **noise** shocks changes interpretation of macroeconomic fluctuations.
- ▶ Real-time information sets matter for forecasting and policy evaluation.

TANK

A parsimonious split between unconstrained and hand-to-mouth households introduces heterogeneous marginal propensities to consume.

HANK

A distribution of households across income, wealth and liquidity positions allows policy to operate through distributional as well as aggregate channels.

Heterogeneity can fundamentally change monetary and fiscal transmission.

1. Choose the economic question and relevant frictions.
2. Write optimization problems and equilibrium conditions.
3. Compute the steady state.
4. Solve / approximate the dynamic model.
5. Check determinacy and identification.
6. Calibrate or estimate parameters.
7. Validate against moments, IRFs and historical dynamics.
8. Perform policy experiments and robustness analysis.

1. DSGEs connect macro dynamics to optimizing behavior and equilibrium.
2. RBC models provide the core solution machinery; New Keynesian models add policy-relevant frictions.
3. Blanchard–Kahn conditions are central for rational-expectations determinacy.
4. Bayesian methods provide a coherent framework for DSGE estimation and uncertainty.
5. Financial frictions, imperfect information and household heterogeneity extend the framework toward modern policy questions.

- ▶ Derive the steady state of the baseline RBC model.
- ▶ Verify the Blanchard–Kahn conditions under alternative Taylor-rule coefficients.
- ▶ Recompute the technology-shock IRFs under different ρ_A values.
- ▶ Compare calibration with Bayesian estimation for selected parameters.
- ▶ Remove one Smets–Wouters friction and compare model fit.
- ▶ Add a financial-friction or heterogeneous-agent extension.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 6. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add derivations of household and firm optimality conditions.
- ▶ Add Dynare, Julia, Python or MATLAB demonstrations.
- ▶ Add your own calibration and estimation exercises.
- ▶ Add policy counterfactuals, alternative shocks and heterogeneous-agent examples.

Make them yours.

Forecasting

From structural models to predictive performance

Next question

How should macroeconomic forecasts be constructed, evaluated, compared and combined?